SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO2 - ASSESSMENT TOOL - 3

ERROR ANALYSIS TASK

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO2**	**Assessment**	**Assessment Tool 3: Error Analysis Task**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO2 Statement	Design effective test cases using black-box, white-box testing techniques and to control methods flow		
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I, M SANDHYA (192312421), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M SANDHYA

ANALYSIS OF ERROR GUESSING AND IMPROVED TESTING STRATEGY FOR A SAFETY-CRITICAL BRAKING SYSTEM

1. Introduction

A braking system is one of the most important safety-related systems in a vehicle. Its main purpose is to reduce vehicle speed or bring the vehicle to a controlled stop when the driver applies the brake or when an automatic emergency-braking function is activated. A defect in this system can cause loss of vehicle control, property damage, serious injury or loss of life. Therefore, the testing process must be systematic, repeatable and supported by sufficient evidence.

In the given scenario, an engineer uses Error Guessing as the primary testing technique and ignores structured black-box testing methods. Error Guessing is useful because an experienced tester can anticipate likely mistakes based on previous defects and practical knowledge. However, it is not sufficient as the main strategy for a high-risk module. It may overlook important input limits, operating states and sequences of events.

This report analyses the limitations of depending only on Error Guessing, explains its effect on test adequacy and proposes a combined strategy using Boundary Value Analysis (BVA) and State Transition Testing. Error Guessing is retained as a supporting technique, while structured methods provide planned and measurable coverage.

1.1 Objectives

Identify the weaknesses of using Error Guessing as the primary technique for braking-system testing.

Explain how incomplete testing can affect defect detection, coverage and safety.

Apply Boundary Value Analysis to important braking-system input limits.

Apply State Transition Testing to normal, braking, emergency and fault conditions.

Justify how the combined strategy improves safety assurance.

1.2 Scope

The discussion focuses on black-box functional testing of braking behaviour, including pressure-related inputs, brake commands, emergency events and sensor-failure conditions. The numerical pressure range used in the examples is illustrative only. In an actual vehicle, valid limits and expected responses must be obtained from approved engineering requirements, component specifications and hazard analysis.

2. Understanding Error Guessing

2.1 Meaning of Error Guessing

Error Guessing is a test-design technique in which the tester uses experience, intuition, knowledge of the application and awareness of common programming mistakes to predict where defects may occur. The tester then designs test cases to expose those possible defects. For example, a tester may try an empty input, an unusually large value, repeated brake commands, a disconnected sensor or a sudden change in input.

The technique is valuable during exploratory testing and can reveal defects that were not explicitly considered in the original test cases. It is also useful after structured test cases have been prepared because it encourages testers to think about unusual and unexpected situations.

2.2 Examples of Error Guessing in a Braking System

Apply the brake repeatedly in a short period to check whether the system responds consistently.

Disconnect or simulate a faulty pressure sensor to observe whether a fault is detected.

Provide an unusually low or high sensor reading and observe the system response.

Release the brake suddenly after an emergency-braking event.

Check whether warning indicators behave correctly when a sensor signal is missing.

Try rapid changes between acceleration and braking commands to identify timing-related defects.

2.3 Why It Should Not Be the Primary Method

The main weakness is that test cases depend on what an individual tester happens to imagine. Two testers may create very different test sets, and neither set may cover all important conditions. There is no guarantee that every critical boundary, state or transition has been examined. Test execution may also become inconsistent when the testing is repeated by another person.

For a safety-critical system, the testing strategy must demonstrate which requirements were tested, what input conditions were used, what results were expected and whether the results passed. Error Guessing alone does not provide a sufficiently structured basis for this evidence.

2.4 Appropriate Role of Error Guessing

Error Guessing should be used as a complementary technique. After BVA and State Transition Testing establish planned coverage, testers can use experience to add unusual scenarios such as intermittent sensor signals, repeated commands, unexpected timing and combinations of faults. This gives the project both systematic coverage and practical exploratory insight.

3. Errors in Relying Solely on Ad-hoc Testing

3.1 Lack of Systematic Coverage

Ad-hoc testing does not ensure that every important input range or operating condition is covered. A tester may check a normal braking operation but fail to test behaviour close to the minimum or maximum permitted pressure. Such omissions are important because defects often occur at limits rather than at ordinary values.

3.2 Dependence on Individual Experience

The quality of Error Guessing varies according to the tester's skill and previous experience. A new tester may not know which sensor failures, timing conditions or event sequences are most dangerous. Even an experienced tester can overlook a condition. The test strategy therefore becomes dependent on individual judgement instead of defined requirements.

3.3 Missing State and Sequence Coverage

A braking system changes behaviour according to its current state. It may be in Normal, Braking, Emergency Braking or Fault Mode. Testing only individual actions without checking sequences can miss defects. For example, the system may work correctly when the brake is pressed from Normal state but fail when an emergency event occurs while ordinary braking is already active.

3.4 Poor Repeatability and Traceability

Ad-hoc test cases may not record exact preconditions, input values, event sequences and expected results. This makes it difficult for another tester to repeat the test and confirm the result. It also becomes difficult to link test evidence to safety requirements and to identify which requirements remain untested.

3.5 Possible Defects Left Undetected

Potential defect	**Possible consequence**
Incorrect pressure limit handling	Insufficient or excessive braking response
Delayed brake activation	Longer stopping distance
Sensor signal failure not detected	Incorrect control decision
Invalid state transition	Unexpected braking behaviour
Failure to recover safely	System remains unavailable or unsafe

These examples show why a high-risk module needs planned test design, documented expected results and review of test coverage. The consequences are potentially severe, so relying only on intuition is not an acceptable basis for safety assurance.

4. Impact on Test Adequacy and Safety

4.1 Meaning of Test Adequacy

Test adequacy refers to whether the selected test cases are sufficient to evaluate the system against its requirements and relevant risks. Adequacy is not measured only by the number of test cases. It also depends on whether important input classes, boundary conditions,

states, transitions, failure conditions and expected outputs have been covered.

4.2 Reduced Defect Detection

If the engineer uses only Error Guessing, test cases may concentrate on familiar or obvious problems. Less obvious defects, such as an off-by-one error at a pressure threshold or an incorrect transition after a sensor fault, may remain undetected. A system can therefore appear to pass testing even though important defects are still present.

4.3 Incomplete Requirement Coverage

Safety-related requirements often specify acceptable input ranges, response times, operating modes, fault reactions and recovery conditions. Each requirement should be linked to one or more test cases. Without a structured approach, the project may not be able to show that all such requirements have been tested.

4.4 Risk to Reliability and Human Safety

A missed defect in a braking system can have consequences beyond software malfunction. It may affect vehicle stability, stopping distance and the driver's ability to avoid a collision. Testing cannot by itself prove that a system is completely safe, but systematic testing provides stronger evidence that specified behaviours and foreseeable fault conditions have been examined.

4.5 Comparison of Testing Approaches

Aspect	Error Guessing alone	Structured combination
Test design	Based mainly on intuition	Based on input limits and state models
Coverage	Difficult to measure	Can be mapped to boundaries and transitions
Repeatability	May vary by tester	Test cases can be documented and repeated
Defect focus	Likely or familiar errors	Boundary, sequence and unusual errors
Safety evidence	Limited	Stronger traceability and documented results

4.6 Overall Assessment

The main impact is reduced confidence in test completeness. A small number of successful ad-hoc tests cannot demonstrate that the braking system behaves correctly across its operating range or during abnormal events. BVA and State Transition Testing address two important gaps by checking values near limits and behaviour across state changes.

5. Boundary Value Analysis (BVA)

5.1 Meaning and Purpose

Boundary Value Analysis is a black-box test-design technique that focuses on values at the edges of an input range. It is based on the observation that defects frequently occur near boundaries because of incorrect comparison operators, rounding, range checks or off-by-one mistakes. Instead of selecting only typical values, the tester checks values at, just below and just above the specified limits.

5.2 Application to Braking Pressure

For illustration, assume that a particular braking-pressure input is specified as valid from 20 to 100 units, inclusive. The values in this example are not universal vehicle specifications. The real limits must be taken from the system's approved requirements. The test objective is to verify that the system handles values at and around the limits according to its defined safety behaviour.

Test case	Input value	Condition	Expected result
BVA-01	19	Just below minimum	Out-of-range condition detected; defined safe response
BVA-02	20	At minimum	Minimum boundary

			handled as specified
BVA-03	21	Just above minimum	Valid input handled correctly
BVA-04	60	Typical middle value	Normal response according to requirements
BVA-05	99	Just below maximum	Valid input handled correctly
BVA-06	100	At maximum	Maximum boundary handled as specified
BVA-07	101	Just above maximum	Out-of-range condition detected; defined safe response

5.3 What the Tester Should Observe

Whether the input is classified correctly as valid or out of range.

Whether the braking response follows the specified control requirement.

Whether a warning or diagnostic indication is produced when required.

Whether the system avoids an unsafe response to invalid sensor data.

Whether the result is consistent when the same boundary test is repeated.

5.4 Importance of Expected Results

A test case must not simply state that an input is accepted or rejected. The expected response should be defined by the engineering requirements. For a safety-related input, the correct response to an out-of-range value may involve fault detection, a warning, a fallback strategy or another defined safe action. The tester must not invent this behaviour during testing.

6. State Transition Testing

6.1 Meaning and Purpose

State Transition Testing is a black-box testing technique used when a system's output or behaviour depends on its current state and on an event. It checks whether the system moves from one state to another correctly and whether invalid or unexpected transitions are handled safely.

6.2 Main Braking-System States

State	**Description**
Normal	System is operating normally and monitoring brake inputs.
Braking	A normal brake command has been detected and braking is active.
Emergency Braking	An emergency condition has triggered the specified emergency response.
Fault Mode	A fault has been detected and the defined fault-handling behaviour is active.

6.3 Example State Transitions

Current state	**Event**	**Expected next state**
Normal	Brake pressed	Braking
Braking	Brake released	Normal
Normal	Emergency condition detected	Emergency Braking

Braking	Emergency condition detected	Emergency Braking, as specified
Braking	Critical sensor fault detected	Fault Mode / defined safe response
Emergency Braking	Emergency condition cleared	State defined by recovery requirements
Fault Mode	Fault cleared and safety checks passed	Normal, if permitted by requirements

6.4 Invalid Transition Testing

The tester must also examine events that should not cause an unsafe transition. For example, the system should not return to Normal merely because a sensor signal briefly changes if the fault has not been cleared and the required safety checks have not passed. Similarly, a reset command should not bypass a safety condition. The exact permitted transitions must be defined by the system requirements and safety design.

6.5 Benefits

Checks behaviour across complete event sequences, not just isolated inputs.

Identifies missing, incorrect or unexpected transitions.

Tests fault detection and recovery behaviour.

Improves coverage of normal and abnormal operating modes.

Creates repeatable test cases from a documented state model.

7. Integrated Testing Strategy: BVA and State Transition Testing

7.1 Why Combine the Techniques?

BVA and State Transition Testing examine different but complementary aspects of the braking system. BVA concentrates on the values supplied to the system, especially values near specified limits. State Transition Testing concentrates on the sequence of events and the operating state in which those events occur. Using both helps identify defects that either technique alone may miss.

7.2 Proposed Testing Workflow

Step	**Activity**	**Output**
1	Review braking requirements and hazard-related conditions.	Approved input limits, states and expected behaviour
2	Identify numeric and categorical inputs.	Input list and valid ranges
3	Design BVA cases around each important limit.	Boundary test-case set
4	Create the state model and list permitted transitions.	State-transition table
5	Design tests for normal, emergency and fault sequences.	Transition test-case set
6	Execute tests and record actual results.	Pass/fail evidence and defect reports
7	Add Error Guessing scenarios and regression tests.	Additional exploratory and repeat tests
8	Review coverage and unresolved risks.	Test adequacy and safety evidence report

7.3 Example of Combined Testing

Consider a situation in which the braking system receives a pressure reading at the upper permitted boundary while the vehicle is already in the Braking state. BVA checks whether the boundary value is processed correctly. State Transition Testing checks whether

the system remains in the correct state and responds correctly to any event that occurs. A combined test can therefore examine both the numeric limit and the system's state-dependent behaviour.

7.4 Sample Integrated Test Cases

Test ID	Initial state / input	Event or value	Expected result
INT-01	Normal; pressure input	Value at minimum limit	Boundary handled as specified; correct state maintained
INT-02	Braking; pressure input	Value just above maximum	Out-of-range handling; no unsafe transition
INT-03	Normal	Emergency event	Transition to Emergency Braking
INT-04	Braking	Critical sensor fault	Fault response and defined safe behaviour
INT-05	Fault Mode	Reset before fault clearance	Reset must not bypass safety requirements

8. Execution, Defect Reporting and Safety Assurance

8.1 Test Execution

Before execution, the test team should confirm the approved requirements, test environment, instrumentation, software version and safety precautions. Tests involving a braking controller should initially be performed in an appropriate simulation, bench or controlled test environment. Vehicle-level testing must follow the organisation's approved safety procedures and be performed by qualified personnel.

8.2 Recording Test Evidence

For every test, the tester should record the test ID, requirement reference, initial state, input value, event sequence, expected result, actual result, pass/fail status, software version and relevant logs. This information makes the test repeatable and supports review of defects.

8.3 Defect Reporting

When actual behaviour differs from the expected result, the tester should create a defect report. The report should include clear reproduction steps, input values, system state, event sequence, observed behaviour, expected behaviour, severity, supporting logs and the related requirement. A potentially safety-critical defect should be escalated through the project's defined safety process.

8.4 Regression Testing

After a defect is corrected, the original failing test must be repeated. Related BVA tests and state-transition tests should also be rerun to check that the correction has not introduced another problem. Regression testing is especially important when changes affect shared control logic, sensor processing or fault handling.

8.5 Measures of Test Adequacy

Requirement coverage: proportion of relevant requirements linked to executed tests.

Boundary coverage: whether important minimum, maximum and adjacent values have been tested.

State coverage: whether each defined state has been exercised.

Transition coverage: whether each required transition has been tested.

Fault-condition coverage: whether defined sensor and system faults have been examined.

Defect status: number and severity of open, fixed and retested defects.

8.6 Safety Assurance

The combined approach increases confidence because it provides documented evidence about input limits, state behaviour and fault handling. However, passing these tests does not alone prove the entire vehicle is safe. Safety assurance also depends on correct

requirements, design reviews, hazard analysis, verification at other levels, hardware validation and compliance with applicable automotive safety processes and standards.

9. Justification, Recommendations and Conclusion

9.1 Justification for Increased Safety Assurance

The proposed strategy is stronger than relying on Error Guessing alone because it replaces informal coverage with planned test conditions. BVA examines values most likely to expose range-checking and threshold defects. State Transition Testing examines whether the system responds correctly as it moves between normal operation, braking, emergency operation and fault handling. Error Guessing remains useful for adding practical scenarios that may not be obvious from the formal models.

Technique	Main contribution to safety assurance
Boundary Value Analysis	Checks critical input limits and values immediately around them.
State Transition Testing	Checks correct behaviour across operating states and event sequences.
Error Guessing	Adds experience-based tests for unusual or unexpected conditions.

9.2 Recommendations

Do not use Error Guessing as the only or primary test-design method for safety-critical braking functions.

Derive all boundary values from approved technical requirements rather than assumed values.

Create and review a complete state-transition model, including fault and recovery behaviour.

Link every test case to one or more requirements and record objective test evidence.

Use simulation and controlled test environments with approved safety procedures.

Repeat relevant tests after every correction and maintain regression coverage.

Review remaining risks with the engineering and safety teams before release.

9.3 Conclusion

Relying solely on Error Guessing is inadequate for a safety-critical braking system because it depends on individual experience and cannot guarantee systematic coverage of important input boundaries, operating states and event sequences. This can allow serious defects, such as incorrect pressure-limit handling, delayed braking, sensor-fault response problems and invalid state changes, to remain undetected.

A combination of Boundary Value Analysis and State Transition Testing provides a more structured and repeatable strategy. BVA verifies behaviour at minimum, maximum and nearby input values, while State Transition Testing verifies the system's response to events in different operating states. Error Guessing can then be used as an additional technique to explore unusual scenarios. Together, these methods improve test adequacy, defect detection, traceability and confidence in the braking system's specified safety behaviour.

References

ISTQB, Certified Tester Foundation Level (CTFL) Syllabus – test design techniques and black-box testing concepts.

ISO 26262, Road vehicles – Functional safety (applicable parts and editions as required by the project).

Project-specific braking-system requirements, interface specifications and approved safety analysis documents.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand